

Supriya Karan

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INFORMATION TECHNOLOGY / SOFTWARE Engineering

Senior QA Engineer

Seasoned Senior QA Engineer with extensive experience leading end-to-end quality initiatives for enterprise-grade endpoint security products at **Broadcom** (formerly **VMware Carbon Black**). Proven expertise in validating advanced security features, ensuring compliance, and maintaining performance and stability across complex Windows environments. Known for elevating product quality through cross-functional collaboration, rigorous testing strategies, and continuous process optimization. Highly skilled in protecting the reliability of mission-critical cybersecurity solutions where even minor defects can disrupt system integrity or introduce real security risks. **I don't just test software—I ensure customers can trust it**, engineering teams can build with confidence, and the business delivers secure, dependable products at scale.

SKILLS

Process / Methodology: SDLC, STLC, Agile, Scrum, DevOps collaboration, CI/CD workflows

Programming Languages: Python, Java (basic)

Test Automation & Management Tools: Pytest, Jenkins, Git (GitLab/GitHub), JIRA, Xray, qTest, TestTrack, Postman, REST APIs

Testing & QA Expertise: Manual & automated testing (Windows, Linux, Mac) UI/Backend Testing, Kernel-level Testing (Windows), API Testing, Interoperability Testing, Debugging, Risk-Based Testing, Defect Tracking, Performance Testing

Compliance & Certification: WHQL Test Kits, Common Criteria (ISO/IEC 15408), Keyboard Accessibility Testing

Cloud & Virtualization: Azure, GCP, VMware VMs

Networking / Protocols: TCP/IP, Wireshark, Client/Server, Active Directory

Soft Skills: Teamwork, Communication, Mentorship, Conflict Management, Analytical Thinking, Stress Management, Leadership, Intercultural Competence, Problem-Solving, Time-Management

AI & Automation Assist: ChatGPT, Jira AI Assist

EXPERIENCE

Broadcom (formerly **VMware Carbon Black**)

July 2008 – Present

R & D Engineer Software 4 (Senior QA Engineer / QA Lead)

Key Achievements & Responsibilities : Contributed significant impact to the Carbon Black App Control product by building comprehensive test suites from the ground up, improving end-to-end test coverage, and enhancing the reliability of complex security functionality. Strengthened release quality through optimized regression workflows, improved defect lifecycle management, and rigorous validation across kernel-level components, drivers, and interoperability with enterprise platforms. Applied deep technical and process expertise to streamline testing operations, reduce execution time, and ensure consistent, compliant, and stable product releases.

Leadership & Process Impact

- Built the product's **entire test suite from scratch**, developed **5,000+ new test cases** and reviewed **7,000+** for accuracy and consistency.
- Led and streamlined QA workflows for **Hotfix, Minor, and Major releases**, ensuring efficient and reliable delivery.
- Improved QA efficiency by restructuring test cases and optimizing processes, achieving a **15–20% reduction in regression time**.
- Led **QA gap analysis**, driving corrective and preventive actions to improve product quality and compliance.
- Optimized the **defect lifecycle**, simplifying reproduction steps to reduce debugging time and accelerate issue turnaround.
- Trained and onboarded new QA engineers, improving team ramp-up and maintaining consistent quality standards.
- Served as **Scrum Master**, facilitating sprints, stand-ups, and retrospectives to enhance team productivity.
- Developed **QA strategies and test plans**, defining scope, approach, and quality objectives for each release.
- Monitored and managed deployments in **production-like environments**, ensuring end-to-end release reliability.

- Collaborated with **DevOps** to assess product health and identify automation opportunities to reduce release risks.
- Successfully migrated **5,000+ test cases from TestTrack to Xray**, improving traceability and overall test management.

SQA roles and responsibilities:

- Optimized **Smoke, Acceptance, and Regression** test suites to reduce execution time and increase test effectiveness.
- Executed **manual** and **automated** testing (functional, regression, system, integration) using **Jenkins CI/CD**.
- Performed **UAT** and **kernel-level testing**, analyzing kernel logs and validating security enforcement modules (execution blocking, process injection prevention, memory access controls, tamper protection).
- Conducted **Common Criteria (ISO/IEC 15408)** compliance testing to meet regulatory and security standards.
- Executed **WHQL / Windows HLK & HCK** testing to validate Microsoft-signed drivers and maintain certification compliance.
- Performed **driver stability testing** using Driver Verifier to detect memory leaks, deadlocks, and other critical issues.
- Conducted interoperability testing across various **Carbon Black** products, streamlining cross-product validation.
- Managed test environment setup, configuration, and maintenance to support reliable QA operations.
- Performed integration testing of **App Control Server** with **Splunk, Syslog, BigFix, and McAfee ePO** for seamless interoperability.
- Developed **negative test scenarios** to detect edge-case failures and improve application robustness.
- Reviewed automated test cases written in **Python**, ensuring coverage and alignment with requirements.
- Used **REST APIs** for manual testing with Postman.
- Conducted **keyboard accessibility checks** to ensure UI compliance with accessibility standards.
- Utilized **Git** for version control and test repository management in continuous testing workflows.

Tech stack: JIRA, Xray, qTest, TestTrack, Jenkins, Git, Python, Pycharm, VMware VMs, VMware AppVolume/AppStack, Microsoft SQL Server, SQLite, Debugging, Rest API, Postman, Wireshark, Remote Desktop, PuTTY, WinSCP; AutoFlex, GCP, Splunk, Syslog, BigFix, McAfee ePO, Client/Server, SDLC, Agile/Scrum, HLK, HCK, WHQL, Driver Verifier.

HCL Technologies – Noida, India

April 2002 – April 2008

Test Lead – Clients: Macromedia (acquired by Adobe), Cadbury Schweppes (UK & US), Dixons Store Group (UK)

- Led QA initiatives for multiple global projects, overseeing test strategy, execution, and reporting for web and enterprise applications.
- Managed W3C compliance and accessibility testing for Cadbury Schweppes' websites, enhancing global usability and adherence to standards.
- Directed technical migration and validation of enterprise applications during business separations, coordinating across multiple business units.
- Conducted performance testing with LoadRunner and managed the defect lifecycle, including root cause analysis and remediation.
- Supervised onsite-offshore QA teams, ensuring timely delivery, quality consistency, and effective communication.

Tech stack: Adobe Dreamweaver, Adobe Contribute, Adobe Fireworks, Adobe Flash, Adobe Extension Manager, Watchfire Bobby, Macromedia DreamweaverMX, W3C HTML Validator, Jigsaw CSS Validator, Jaws (Screen Reader), Inspect 32, Accessibility Tool Bars, SQL, HTML/CSS, Java Script, DHTML, XML

Tata Infotech Ltd - Noida, India

July 1998 – April 2002

Software Engineer - Clients: United Airlines, Montana Dept. of Revenue, Artius

- Built and tested secure enterprise applications for crew scheduling, payroll, and tax processing.
- Gained foundation in software engineering, QA methodologies, and client collaboration

Tech stack: Java 2.0(JSP, JDBC, SWINGS), HTML, Java Script, MS-Windows 98, Rational Rose, Pro*C;

EDUCATION

B.E. (Mechanical) – Birla Institute of Technology, Mesra, India | First Class

Certifications: Sun Certified Java 1.4

Training record: 6 Months Training (at Tata Infotech Ltd): Covered programming (C, C++, Java, FORTRAN), databases (Oracle, RDBMS), GUI/web tools (UML, VB, VC++, PowerBuilder, HTML), software engineering principles, testing, project management, quality practices, team collaboration and Leadership.